



Microwave-assisted grafting of dual-backbone guar gum/sodium alginate superabsorbent polymer: mechanistic insights into sustainable swelling kinetics and soil amendment potential

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Abstract

Implementing soil amendments is a viable strategy to alleviate the negative effects of drought stress. Superabsorbent polymers (SAPs) are promising soil amendments for mitigating drought stress by enhancing soil water retention and availability. In this study, a novel dual-backbone SAP composed of guar gum and sodium alginate grafted with acrylamide (GG/NaAlg-g-Am) was synthesized via microwave-assisted graft polymerization. Reaction parameters—including monomer, initiator, crosslinker concentrations, microwave power, and reaction time—were optimized to maximize grafting efficiency. Successful grafting and microstructural features were confirmed through FTIR and SEM analyses. Swelling studies revealed pH-dependent behaviour, with the highest equilibrium swelling (~ 900 g/g) at neutral pH 7. Kinetic modelling indicated zero-order and Korsmeyer–Peppas model fits, suggesting sustained swelling through controlled fluid diffusion and matrix erosion. Soil amendment trials with 1% GG/NaAlg-g-Am demonstrated a 190% increase in water holding capacity, 32% increase in porosity, and 28% enhancement in soil organic matter, alongside biodegradability with a 110-day half-life. These results highlight the potential of GG/NaAlg-g-Am as a high-performance, sustainable SAP for soil conditioning applications. The aforementioned results validate the efficacy of GG/NaAlg-g-Am superabsorbent polymer as a soil amendment.

Keywords Graft polymerization · Microwave irradiation · Swelling kinetics · Mathematical models · Soil properties · Biodegradability

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Introduction

As the global population continues to rise, agriculture worldwide is confronted with numerous challenges, including climate change, urban expansion, the sustainable management of natural resources, runoff losses, and the buildup of pesticides and fertilizers. A significant concern for agriculture is global water scarcity, as drought represents a natural disaster that impacts farmers and the broader socio-economic conditions of their countries. Increased urban activities, regular constructions, and inadequate waste disposal management stir up the defilement of soil properties [1]. Poor agricultural practices like long-term cultivation, overgrazing, deforestation, monocropping, burning of crop residues and overreliance on pesticides and chemical fertilizers drive soil erosion and thus decrease soil properties [2]. Depletion in soil properties like water holding capacity, porosity, particle density and soil organic carbon results in a decrease in the yield of plant crop. A requirement arises to enhance soil properties by external means in the form of soil amendment [3, 4]. Consequently, formulating agronomic practices aimed at reducing drought stress in cultivated crops is paramount. Many research studies have been carried out on various types of soil amendments, highlighted ones among the list are manure, wood products (sawdust), biosolids like byproducts of sewage treatment and compost. The aforementioned ways of amendments exhibit their cons, like fresh manure can harm plant growth due to its elevated ammonia level. Wood products tie up nitrogen in the soil and result in nitrogen deficiency in plants. The primary concerns associated with biosolids are the presence of heavy metals, salts, and pathogenic microorganisms [4]. The duration necessary for these amendment materials to enhance the soil properties is excessively long. To address these limitations, the practice of treating soil with organic polymer materials emerged [5]. Many synthetic polymers that can form hydrogels, like polyacrylamide [6], polyvinyl alcohol, polyacrylonitrile, etc., are used as soil amendments [7].

Hydrogels are a 3D polymeric network that can retain large quantities of water in the network space. This water absorption or retention is due to the presence of -OH, -NH₂, -COOH, -SO₃H and many more functional groups within the molecular structure [8, 9]. In 1952, the research on soil conditioning with synthetic polyelectrolytes was introduced with various synthetic polymers to improve soil properties such as aeration, aggregation and water-holding capacity [10]. Koslow worked on a higher molecular weight polymer, poly (ethene oxide), as a soil stabilizer [11]. In order to improve the efficiency and longevity of the polymer treatment, the crosslinking of the polymers was introduced. Cooke synthesized a methylene bisacrylamide crosslinked polyacrylamide polymer added to the plant growth media, where a significant effect on seedling emergence is observed. The synthetic polymers yielded positive results in soil conditioning, but they led to degradation issues of polymers, adversely impacting soil quality and crop productivity [12].

Natural polymers like cellulose, starch, Xanthan gum, chitosan, etc., were utilized as alternative sources of soil conditioner [13–15]. Shulga et al. developed lignin-based interpolymer complexes that act as an adhesive to soil and increase soil stability and decrease erosion [16]. Wang et al. studied the effect of hydrophilic polymers such as starch-based, polyacrylamide-based and propenoate-propanamide copolymer on the

soil properties, where the water retention capacity of starch-based polymer composite was higher than the other two composites [17]. The study has been conducted to enhance water retention capacity while simultaneously achieving controlled release of fertilizer through the synthesis of cellulose acetate-coated fertilizer [18, 19].

The present study focuses on the natural polymer guar gum and sodium alginate. Guar gum is a natural polymer that has galactomannan polysaccharide extracted from the guar bean. The backbone consists of a linear sequence of β 1,4-linked mannose residues, with galactose residues linked at the 1,6 positions on every alternate mannose, resulting in the formation of short side branches. This substance is soluble in water and can form a gel when crosslinked, enhancing its suitability as a material for the synthesis of hydrogels [20]. Guar gum-based composites and hydrogels were studied on application as a soil conditioner to increase water absorption and retention capacity and also controlled release [21]. Sodium alginate is derived from the cell wall of brown algae. It is a linear polysaccharide comprised of 1,4- β -D-mannuronic (M) and α -L-guluronic (G) acids [22]. It is used as an emulsifier, stabilizer and thickening agent. According to the Food and Agriculture Organization (FAO), global seaweed production amounted to 14.7 million tonnes in 2005; by 2015, it had doubled to 30.4 million tonnes over ten years [23]. Sodium alginate is regarded as a key polymer in agricultural applications due to its abundant availability and notable properties such as swelling and gelling capabilities. Idrissi et al. have researched the impact of sodium alginate-based hydrogels on the growth of tomato plants in conditions of water scarcity [24].

Recently, many hydrogels have been synthesized and applied as soil conditioners. Alessandro Sorze et al. have studied the effect of cellulose fillers on the properties of xanthan-based composite as soil conditioners, where 1.8% of polymer treatment improved soil water absorbing capacity from 34% to 64% [25]. There is a lack of sufficient research concerning tailoring two backbone polymers with a monomer via the microwave irradiation method and its application as soil conditioning. The current study focuses on synthesizing guar gum and sodium alginate backbone polymers grafted with acrylamide monomer via microwave irradiation method. The effect of the guar gum to sodium alginate ratio, the concentration of monomer, crosslinker and initiator, reaction time and microwave power level were evaluated. The soil was treated with GG/NaAlg-g-Am superabsorbent polymer and its impact on soil particle density, porosity, maximum water holding capacity, and water retention capacity was assessed. The investigation into the biodegradability and swelling kinetics of the GG/NaAlg-g-Am superabsorbent polymer has highlighted its potential benefits for its use as a soil conditioner.

Thus, the novelty of this work lies not only in the dual-backbone graft copolymer synthesis of guar gum and sodium alginate but also in the comprehensive, mechanistically guided approach that integrates microwave-assisted synthesis, sustainable swelling kinetics, structure–function correlation, and practical soil application validation. Unlike previous studies using single polysaccharide systems, this dual-polymer strategy enhances the mechanical strength, porosity, and water retention capacity of the resulting GG/NaAlg-g-Am hydrogel. The hydrogel achieved high equilibrium swelling (~ 900 g/g), enabled by strong C–C covalent crosslinks via MBA and hydrogen bonding between functional groups, ensuring controlled, sustained swelling

despite its porous structure. Multi-model kinetic analysis and SEM-supported structural insights further confirmed its superior performance. These combined innovations provide a technical advancement in superabsorbent polymer design and offer novel solutions for soil conditioning and environmental remediation, setting this work apart from existing literature.

Materials

Guar gum powder (CAS No. 9000-30-0) and sodium alginate 91% (CAS No. 9005-38-3) were procured from Loba Chemie Pvt. Ltd.; both are non-hazardous and not classified under ADR transport regulations. Acrylamide 98% (CAS No. 79-06-1; ADR Class 6.1 – toxic substances, UN 2074, PG III), acetone 99% (CAS No. 67-64-1; ADR Class 3 – flammable liquids, UN 1090, PG II), and Ammonium persulfate 98% (CAS No. 7727-54-0; ADR Class 5.1 – oxidizing substances, UN 1444, PG III) were also sourced from Loba Chemie Pvt. Ltd. N, N'-Methylenebisacrylamide AR grade (CAS No. 110-26-9) and sodium hydroxide pellets AR grade (CAS No. 1310-73-2; ADR Class 8 – corrosive substances, UN 1823, PG II) were purchased from S D Fine-Chem Ltd.

Methods

Microwave-assisted polymerization of GG/NaAlg-g-Am superabsorbent polymer

The design protocol for the synthesis of GG/NaAlg-g-Am superabsorbent polymer is based on prior research that involved the microwave-assisted synthesis of guar gum grafted with acrylic acid superabsorbent polymer [26]. The combination of sodium alginate and guar gum in a 7:3 ratio is formulated using distilled water. A solution containing 31 wt% acrylamide, which serves as a monomer, is introduced into the reaction mixture while stirring to ensure the uniformity of the solution. Ammonium persulfate was added as an initiator at a concentration of 1.6 wt%, along with 3.8 wt% of N, N'-Methylenebisacrylamide as a crosslinker, and the mixture was stirred continuously for 10 min. The reaction mixture underwent microwave irradiation for 3 min at a power level of 720 W. The resulting grafted polymer solution was adjusted to a pH of 7 using 0.1 N NaOH, and subsequently precipitated in acetone to eliminate homopolymers. The precipitate was washed multiple times with acetone and dried at 60 °C in a hot air oven for 6 h until a constant weight was achieved.

Optimization of reaction parameters

A series of GG/NaAlg-g-Am superabsorbent polymers was synthesized by varying the ratios of sodium alginate to guar gum, as well as the concentrations of acrylamide (monomer), Ammonium persulfate (initiator), and N, N'-Methylenebisacrylamide (crosslinker). Additionally, variations in reaction time and microwave power levels

were implemented. The grafting percentage for each trial was determined using Eq. 1 [27].

The reaction parameters have been optimized for the highest grafting percentage and their effects on grafting efficiency were investigated. The concentration of the N, N'-Methylenebisacrylamide crosslinker is optimized at the highest E.S [28, 29]. E.S. was calculated using Eq. 2 [30].

$$\text{Grafting percentage} = \frac{(\text{weight of grafted polymer} - \text{weight of base polysaccharide})}{(\text{weight of base polysaccharide})} \times 100 \quad (1)$$

$$E.S. = \frac{(\text{weight of swollen sample} - \text{weight of dried sample})}{(\text{weight of dried sample})} \quad (2)$$

FTIR spectroscopy

FTIR spectra of pure guar gum, sodium alginate, acrylamide and GG/NaAlg-g-Am superabsorbent polymer were recorded using a PerkinElmer FTIR spectrometer between 400 and 4000 cm^{-1} . The sample was mixed with KBr (1 mg/900 mg) to form pellets for sample preparation before testing.

Scanning electron microscope

The surface morphology of sodium alginate, guar gum, and GG/NaAlg-g-Am superabsorbent polymer was analysed using a scanning electron microscope (TESCAN VEGA). The specimen was gold sputtered before analysis.

Swelling kinetics

A quantity of 0.1 ± 0.001 g of GG/NaAlg-g-Am superabsorbent polymer powder, sized between 100 and 150 mesh, was placed in a test tube and an equivalent volume of distilled water was introduced. The weight of the swollen sample was recorded at 10-minute intervals. This process was continued for a total duration of 180 min. The effect of pH on swelling behaviour is examined through a comparable methodology utilizing various buffer solutions, specifically an acidic medium at pH 4, a neutral medium at pH 7, and a basic medium at pH 10.2 [31].

Soil collection and Preparation

A soil sample was collected from the garden area of B. M. S. College of Engineering campus, and the soil texture was assessed through the analysis of particle size distribution. The soil sample consists of sand 86%, clay 7.5% and silt 5.5% [32]. It has a pH of 6.1 ± 0.3 [29] and moisture content of $12.53 \pm 0.38\%$ [33]. The collected soil was dried in the sunlight.

Particle size distribution of the soil sample

The soil sample collected was subjected to particle size distribution by passing it through various-sized mesh ranging from 5 mesh to 200 mesh. The weight retained on each mesh was weighed. The particle size distribution was analysed with the plot of cumulative mass fraction against average particle size. The soil that passed through a 50 mesh was taken for further studies and application.

Biodegradability of polymer

The soil burial method was applied for biodegradability studies of GG/NaAlg-g-Am superabsorbent polymer [34]. A polymer sample of approximately 100 ± 0.1 g of GG/NaAlg-g-Am superabsorbent polymer was enclosed in a 60-mesh nylon bag. This bag was then buried in the soil at a depth of 5 to 8 cm. Water was introduced to ensure that the moisture content remained at approximately one-third of its water retention capacity. At specific time intervals, the sample was extracted, gently washed to eliminate any adhering soil, and then dried until a constant weight was achieved. The dry weight of the degraded polymer was recorded. The rate constant for decomposition was calculated utilizing Olson's single exponential model, as expressed in Eq. 3 [35]. Percentage degradation is calculated based on dry weight remaining at that particular interval of time using Eq. 4 and the half-life period is calculated using Eq. 5 [36]. Morphological changes due to biodegradation are validated by SEM analysis [37].

Olson model,

$$W_t = W_o e^{-kt} \quad (3)$$

W_t = dry weight of GG/NaAlg-g-Am superabsorbent polymer at time t (g).

W_o = initial dry weight of GG/NaAlg-g-Am superabsorbent polymer (g).

k = degradation rate constant.

$$\% \text{ dry weight remaining} = \frac{\text{Dry weight at time } t}{\text{Initial dry weight}} \times 100 \quad (4)$$

$$\text{Half-life period, } t_{1/2} = \frac{0.693}{k} \quad (5)$$

Agricultural application

Hydrogels play a significant role in agricultural applications by serving as water reservoirs within the soil. They enhance various chemical and physical properties, including porosity, bulk density, and organic carbon content, which underscores their substantial demand as soil amendments [38, 39].

Maximum water holding capacity

A specific quantity of soil is treated with a predetermined amount of 100 mesh-sized GG/NaAlg-g-Am superabsorbent polymer. The GG/NaAlg-g-Am superabsorbent polymer-treated soil (PTS) samples ranged from 0.2% to 1.0%, with increments of 0.2% utilized for subsequent studies. The soil sample without any polymer treatment was considered as a control. The maximum water holding capacity (MWHC) of soil was determined by Hilgard's method [40]. Air-dried soil was compacted into a cylindrical container featuring perforations at the base, which was covered with filter paper. This container was then submerged in a tub filled with excess water, allowing water to permeate the soil sample from the bottom. After a duration of 12 h, the cylindrical container was removed from the tub and weighed once dripping had ceased. The wet soil sample was subsequently dried at a temperature of 110 °C and weighed to determine its dry weight. The moisture holding capacity (MWHC) of all the PTS samples and the control was calculated by directly measuring the difference between the wet and dry weights of the samples [41]. The effect of GG/NaAlg-g-Am superabsorbent polymer treatment on the water imbibition in the soil was analysed.

Soil particle density, bulk density and porosity

The particle density of the soil was determined by the volumetric flask method [42]. Soil was oven-dried at 105 °C till constant weight and sieved through a 2 mm sieve. 25 g of dried soil sample was taken in a volumetric flask. About 50 mL of distilled water was added to the soil flask. It was boiled for 10 min to remove air bubbles. The soil mixture was allowed to cool. Once cooled, the cap was closed and allowed it settle for 24 h. The flask was filled with distilled water up to the mark. The weight of the soil-water mixture was weighed. The temperature was noted. The particle density was determined using the mass by volume ratio of soil, with all the variables tabulated as per Table 1 [43]. Similarly, the particle density of the PTS samples was also determined. All the trials were carried out in triplicate.

The bulk density of soil was determined using standard laboratory methods [44]. The soil sample was collected in the previously weighed cylinder using a core sampling tool. The soil was dried to 105 °C until constant weight. Bulk density was calculated using Eq. 6. The same experiment was carried out in triplicate for other PTS samples. The porosity of soil refers to the proportion of voids present within the soil, expressed as the percentage of pores to solid particles, which is interrelated to particle density and bulk density. It was calculated using Eq. 7 [45].

$$\text{Bulk density} = \frac{\text{weight of (soil + cylinder)} - \text{weight of cylinder}}{\text{Volume of cylinder}} \quad (6)$$

$$\text{Porosity percentage} = \left(1 - \frac{\text{bulk density}}{\text{particle density}}\right) \times 100 \quad (7)$$

Table 1 Regression coefficients of mathematical models across various pH media with their characteristic equations

Model name	Characteristic equation	R^2 value		
		pH4	pH7	pH10.2
Polynomial regression [93]	$E.S.t = a_0 + a_1t + a_2t^2 + a_3t^3 + \dots + a_nt^n$ $a_0, a_1, a_2, \dots, a_n$ $= regression\ coefficient$	0.997	0.991	0.969
Weibull model [94]	$E.S.t = E.S_{max} (1 - \exp(-at^b))$	0.92	0.891	0.84
Logistic model [95]	$E.S.t = \frac{E.S_{max}}{1 + \exp(-k(t-t_0))}$	0.692	0.995	0.756
Gompertz model [96]	$E.S.t = E.S_{max} \exp(-\exp(-k(t-t_0)))$	0.745	0.624	0.78

Water retention capacity

The influence of GG/NaAlg-g-Am superabsorbent polymer on the water retention capacity of soil was investigated utilizing the pressure plate apparatus method (IS 14765 (2000)) across various matric tensions, specifically 0.5 bar, 1 bar, 2 bar, 5 bar, and 10 bar. Both control and PTS at varying concentrations of 0.2%, 0.4%, 0.6%, 0.8%, and 1.0% were placed in rubber rings positioned on ceramic plates and allowed to saturate overnight. Following saturation, the ceramic plates were arranged in a pressure chamber set to different tensions. The apparatus was operated until the water flow from the chamber completely ceased. The dehydrated, moist and final samples were promptly weighed. The moist samples were subsequently dried in a hot air oven at 105 °C for 24 h, allowed to cool in air, and then reweighed [46]. The volume of water retained at specific pressures was determined using Eq. 8 [47].

$$\% \text{ Moisture retention} = \frac{(\text{mass of wet soil} - \text{mass of dried soil})}{\text{mass of dried soil}} \times 100 \quad (8)$$

Soil organic matter (SOM)

The Soil Organic Matter (SOM) content plays an important role in plant growth. The SOM is determined by the standard ASTM-D 2974 [48]. Oven-dried soil was ignited in a muffle furnace at 440 °C overnight. The % SOM was determined for control and PTS using Eq. 9 [49]. The procedure was repeated for a 140-day incubated soil sample and %SOM was compared.

$$\% \text{ Soil Organic Matter} = \frac{(\text{mass of dried soil} - \text{mass of ignited soil})}{\text{mass of dried soil}} \times 100 \quad (9)$$

Results and discussion

Mechanism of polymerization

The proposed polymerization mechanism is depicted in Fig. 1, illustrating the microwave-assisted free-radical grafting of guar gum and sodium alginate with acrylamide in the presence of N, N'-Methylenebisacrylamide as a crosslinker.

Upon microwave irradiation, Ammonium persulfate undergoes thermal decomposition, producing sulphate free radicals ($\text{SO}_4^{\bullet-}$) that abstract hydrogen atoms from the hydroxyl groups of guar gum and sodium alginate. This initiates the formation of macroradical sites on the dual polymer backbones, which then attack the vinyl group ($-\text{CH}=\text{CH}_2$) of acrylamide monomers, initiating free radical polymerization [50]. The resulting polyacrylamide chains are covalently grafted onto the guar gum/sodium alginate backbones, forming a copolymer network and increasing the grafting efficiency. As active sites become depleted via radical termination mechanisms, namely disproportionation, recombination, residual acrylamide may undergo homo polymerization, thereby reducing the graft yield. Meanwhile, the N, N'-Methy-

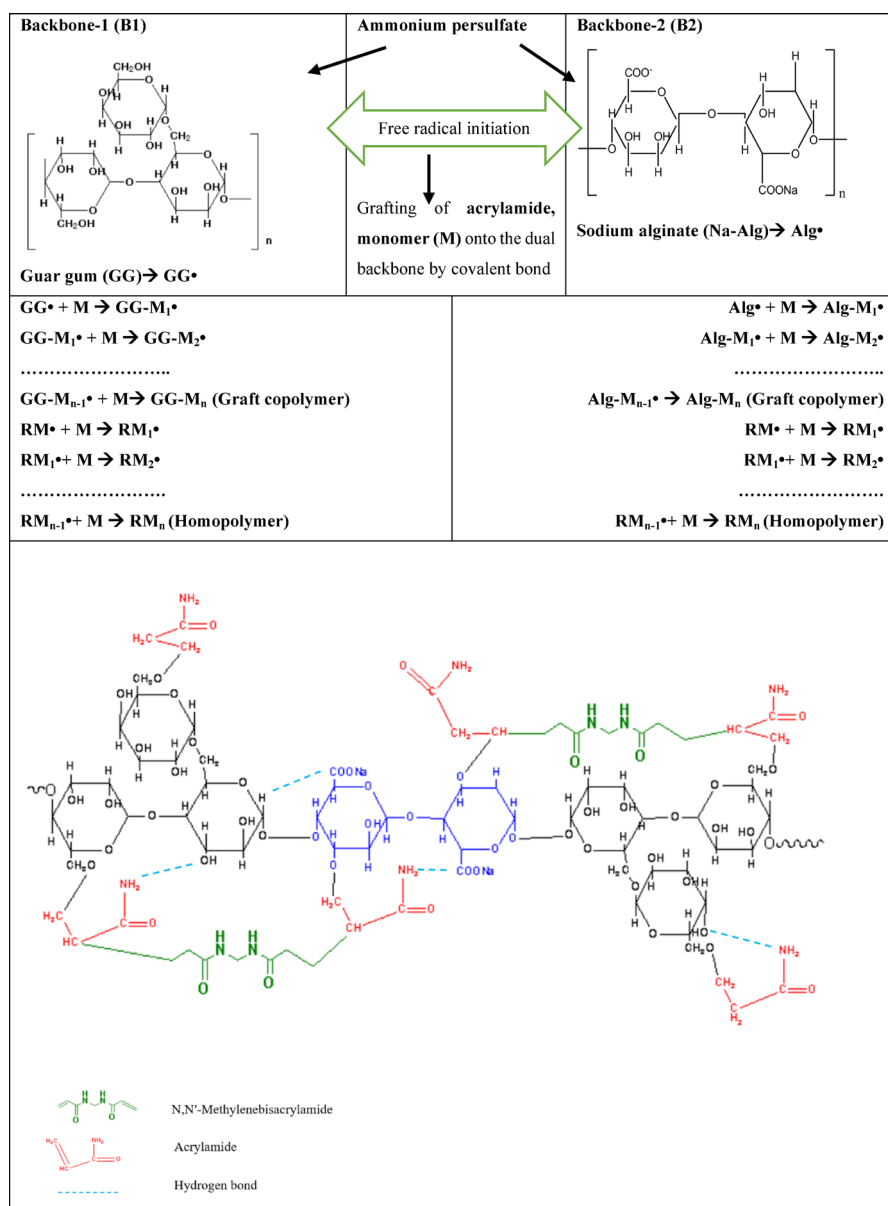


Fig. 1 Mechanistic understanding of polymerization

lenebisacrylamide crosslinker facilitates the formation of three-dimensional covalent bridges ($-\text{NH}-\text{CO}-\text{NH}-\text{CH}_2-$) between polyacrylamide chains and the guar gum/sodium alginate backbones, resulting in a crosslinked hydrogel network [51, 52]. This network is further stabilized by hydrogen bonding among $-\text{OH}$, $-\text{COOH}$, and $-\text{NH}_2$ groups, producing a porous 3D polymer matrix as observed in SEM analysis [53, 54]. The interconnected pores enhance water diffusion and retention, while

the optimal crosslink density imparts mechanical stability and supports sustainable swelling behaviour over extended durations. This structure underpins the hydrogel's performance as a superabsorbent polymer (SAP) with significant potential for soil amendment applications.

Optimization of reaction parameters

Grafting percentage plays an important role in the optimization of reaction parameters. The effect of reaction parameters on grafting percentage was evaluated.

Effect of monomer

Monomer concentration is the key factor in the grafting efficiency. From Fig. 3a, the highest grafting percentage was attained at 31% acrylamide concentration; further increase in concentration resulted in a decrease in grafting percentage. This behaviour was attributed to the formation of copolymers through graft polymerization in the vacant sites on guar gum and sodium alginate matrices, which were generated by Ammonium persulfate initiator and microwave irradiation [55]. Acrylamide consists of two functional groups: amide and alkene. An addition reaction takes place with alkenes through the cleavage of the double bond, which accounts for the initial rise in grafting percentage as the concentration of acrylamide increases [56]. As the radical sites were completely occupied, and a further increase in monomer concentration resulted in the formation of a homopolymer, i.e., an additional chain reaction between the alkene group of acrylamides [57]. This contributes to the decrease in grafting percentage at 37.5% of monomer concentration. A comparable pattern of monomer concentration was noted by Teena et al. in their investigation into the synthesis and swelling characteristics of graft copolymerized isotactic polypropylene film [58].

Effect of the sodium alginate to Guar gum ratio

Guar gum and sodium alginate serve as the backbone for grafting polymerization, and the proportion between them significantly influences the grafting efficiency. The data presented in Fig. 3b indicates that the optimal grafting percentage 248% is achieved at a NaAlg/GG concentration ratio of 2.3 or 7:3. Further increments resulted in a constant grafting percentage. Guar gum backbone is a linear chain of β 1,4-linked mannose residues, at every second mannose is attached to a 1,6-linked galactose side chain. The b carbon unit of the pyranose ring becomes activated through the ring opening as a result of free radical polymerization, allowing it to bind with the Ammonium persulfate initiator and the acrylamide radical. Concurrently, the other carbon transforms into an aldehyde group due to the elimination of a proton, which indicates a successful grafting process [59]. Sodium alginate consists of mannuronic (M) and guluronic acid (G). The mechanism of grafting may be understood such that free radicals are generated following the cleavage of C–H, C–O, and C–C bonds via processes such as hydrogen abstraction, chain scission, and cycle opening. The introduction of acrylamide ensures that no unsaturated bonds are present once a radical is formed. This leads to the formation of a copolymer and contributes to a higher graft-

ing percentage [60]. Once the ratio of guar gum and sodium alginate has saturated with occupied radical sites, i.e., no grafting polymerization occurs, further increase in backbone results in nearly constant grafting percentage.

Effect of ammonium persulfate initiator

Ammonium persulfate acts as an initiator. Figure 3c depicts the effect of varied concentrations of Ammonium persulfate from 1 to 3 wt% on the grafting percentage. Initially, with an increase in Ammonium persulfate concentration till 1.6 wt%, grafting percentage also observes an increasing trend, followed by a decrease in grafting percentage. Ammonium persulfate, when added to the reaction mixture, the dianion dissociates into radicals, as shown in the following chemical Eq. 10. Sulphate radical created adds to the double bond and a propagating radical site was created onto the guar gum and sodium alginate backbone, leading to the additional polymeric chain reaction of monomer onto the backbone. This results in an increase in grafting percentage [61]. Decrease trend of grafting percentage may be attributed to the termination reaction that occurs when propagating radicals of the backbone undergo disproportionation with acrylamide monomer, resulting in the formation of homopolymers [62]. Metin et al. have made comparable observations, noting the use of benzoyl peroxide as an initiator in the synthesis of poly(ethylene terephthalate) fibres that are grafted with N-(hydroxymethyl) acrylamide through a free radical process [63].



Effect of N, N'-Methylenebisacrylamide crosslinker

N, N'-Methylenebisacrylamide serves as a crosslinker, which plays an important role in the swelling ability of GG/NaAlg-g-Am superabsorbent polymer. A series of polymers is synthesized with varied concentrations of crosslinker from 1 to 9 wt%. Figure 3d indicates that maximum swelling capacity is reached around 4 wt% crosslinker concentration, further increase resulted in a decrease in swelling capacity. An increase in swelling capacity can be justified as crosslinking occurs due to radical copolymerization of N, N'-Methylenebisacrylamide with acrylamide, resulting in crosslinked gels and thus a three-dimensional network is formed. In turn highly void structure results in the storage of abundant amounts of water through hydrogen bonds. Further increase in crosslinker concentration from 4 wt% to 9 wt% results in a decline of swelling capacity. This sweep might be owed to extensive crosslinking, leading to increased rigidity and diminished flexibility. This results in the shrinkage of void space in the intermolecular structure and a reduction in the absorbance of water [64–66]. Optimized crosslinking and dense crosslinking are illustrated in Fig. 2. A comparable result was recorded by Haibo et al. during the synthesis of hydrogels using rosin-based crosslinking agent [67].

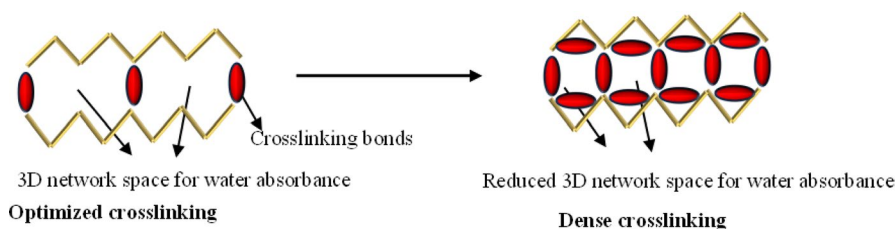


Fig. 2 Effect of optimized crosslinking vs. dense crosslinking on polymer network space

Effect of microwave power

Microwave irradiation remarkably minimizes the usage of toxic solvents and reaction time and improves product yield during graft polymerization [68]. A series of GG/NaAlg-g-Am superabsorbent polymers was synthesized by adjusting the microwave power between 540 and 900 W. As illustrated in Fig. 3e, it is observed that from 540 W to 720 W, the grafting percentage initially increases; however, a further increase in microwave power results in a decline in the grafting efficiency. The initial increase in grafting percentage is attributed to enhanced free radical generation under microwave irradiation, which accelerates the polymerization kinetics. However, beyond 720 W, the grafting efficiency decreases due to premature radical termination and increased homopolymer formation, which impede effective graft copolymerization [69]. At higher energy levels, due to the disproportionation of free radicals on the backbone and monomer, leading to the formation of homopolymers due to pulling hydrogen from another radical chain, this prompts a sequential decrease in grafting percentage [70].

Effect of reaction time

Microwave irradiation plays a significant role in the reduction of reaction time compared to the conventional method of grafting. In Fig. 3f reaction time is varied from 1 to 5 min at an interval of 1 min. The highest grafting percentage is attained at 3 min of reaction time; the progressive increase in time resulted in a decrease in grafting percentage. The results can be elucidated by noting that free radical polymerization takes place initially under microwave irradiation, resulting in an increased grafting percentage as the reaction time extends. However, with further prolongation of the reaction time, the formation of homopolymers occurs due to the overheating effect induced by microwaves, which subsequently causes a reduction in the grafting percentage [71]. Analogous results were reported by Yu et al. in the modification of microcrystalline cellulose with acrylamide under microwave irradiation [72].

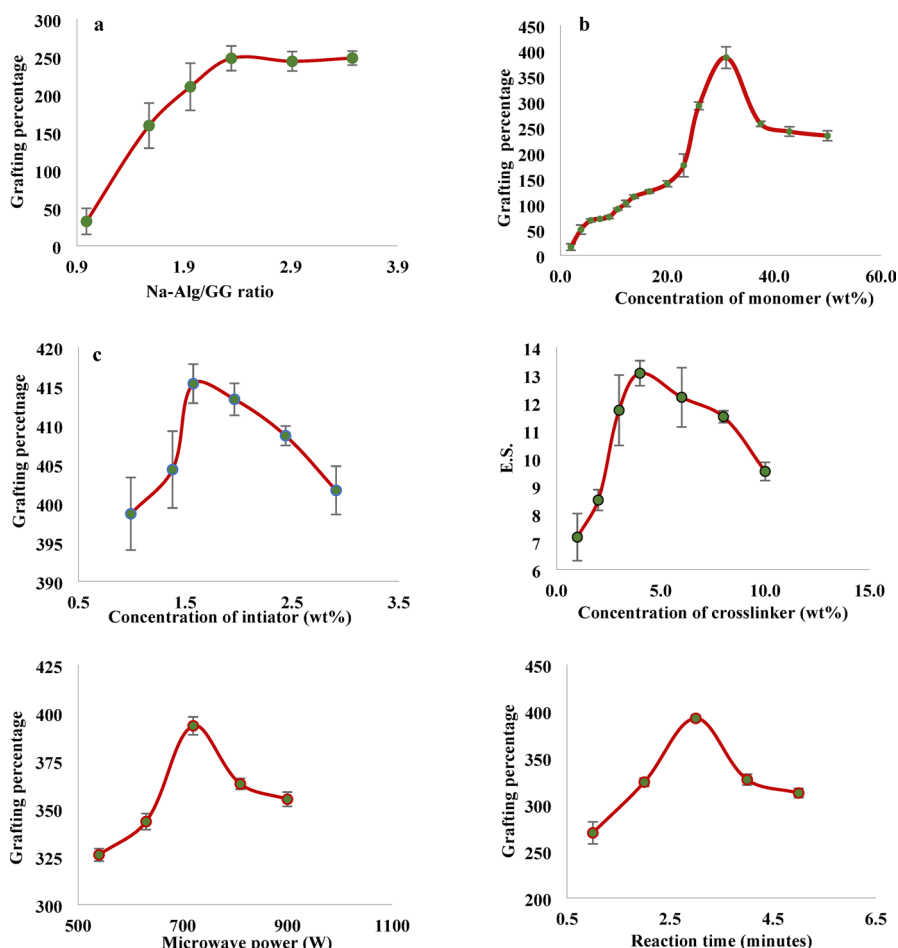


Fig. 3 Effect of reaction parameters on grafting percentage

Characterization

FTIR spectroscopy

Fourier transform infrared spectroscopy of guar gum, sodium alginate and GG/NaAlg-g-Am superabsorbent polymer is depicted in Fig. 4. In the pure guar gum spectrum, the broad spectrum at 3423 and 1019 are the characteristic features of a compound due to stretching vibration of -OH due to hydrogen-bonded OH bond decreasing the absorption frequency and C-O group, respectively. A sharp band at 2925 indicates the C-H stretch of the CH_2 group, i.e., the aliphatic -CH functional group. The band at 1608 is due to ring stretching of the mannose and galactose unit [73]. IR spectra of sodium alginate exhibit a characteristic broadband at 3415 due to the absorption vibration of -OH (hydroxyl groups). A sharp band at 1608 and 1384 indicates the asymmetric and symmetric stretching vibration of the -COO group. The

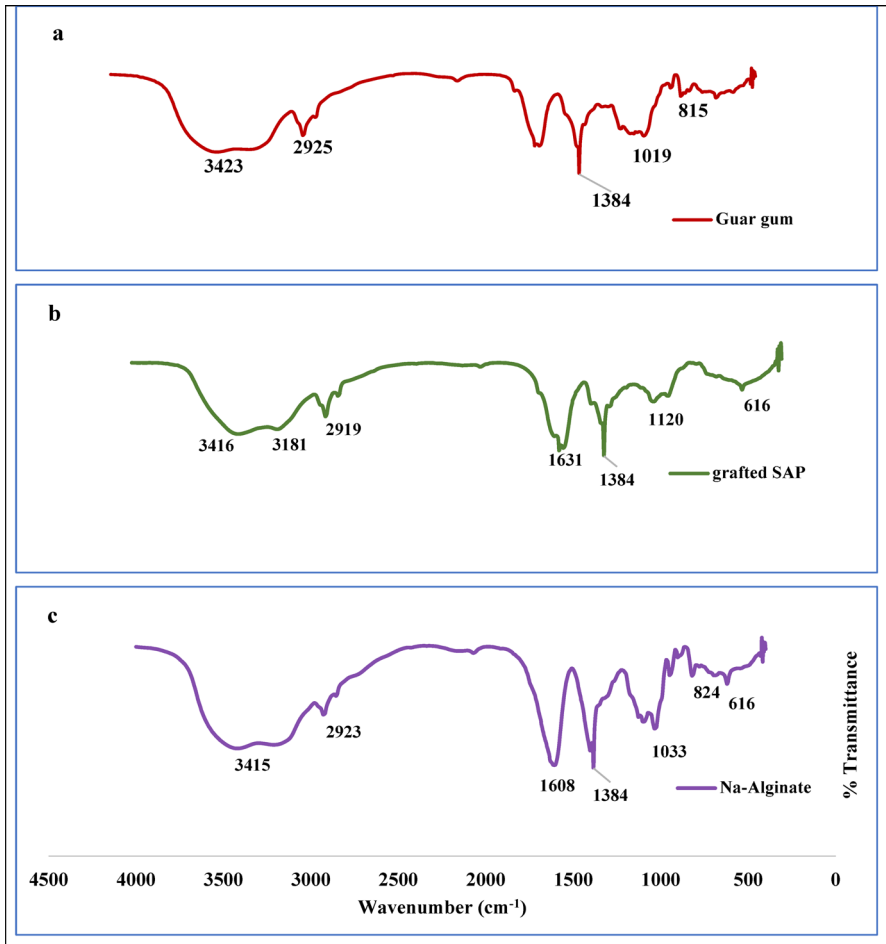


Fig. 4 FTIR spectra of pure guar gum, GG/NaAlg-g-Am superabsorbent polymer and sodium alginate

Narrowband at 1033 represents vibration due to elongation of the C-O group [74]. GG/NaAlg-g-Am superabsorbent polymer IR spectroscopy shows a 3416 broadband due to -OH stretching of the hydroxyl group. A unique weak peak appeared at 3181, indicating the N-H stretching due to the grafting of acrylamide onto guar gum and sodium alginate. The medium peak at 2919 indicates C-H stretching of the alkane group. The sharp peaks at 1631 and 1384 are due to asymmetric vibration of the -COO group, which is slightly deviated compared to guar gum and sodium alginate, confirming grafting of GG/NaAlg-g-Am superabsorbent polymer. A weak peak at 1120 in the fingerprint region may be attributed to H-C-O bond bending due to the crosslinking of acrylamide and guar gum/sodium alginate backbone [75, 76].

SEM

Scanning electron microscopic image of pure guar gum, and sodium alginate shown in Fig 5(a) and (b) respectively are compared with GG/NaAlg-g-Am superabsorbent polymer recorded at 10 μm and 2 μm in Fig. 5(c) and (d) respectively. The characteristic topology of pure guar gum of lamellar texture is noticeable [77]. The surface of sodium alginate exhibited a granular texture. It is visible that the morphology of GG/NaAlg-g-Am superabsorbent polymer has a porous structure, indicating the grafting and crosslinking effect of the polymerization reaction. The images confirm that the GG/NaAlg-g-Am superabsorbent polymer exhibits a porous structure, with the pores likely formed in the hydrogel due to water evaporation caused by the heat generated during the reaction. The pore size variation observed in Fig. 5d suggests the heterogeneous crosslinking of guar gum and sodium alginate regions. Microwave irradiation plays an important role in uniform grafting and crosslinking, thus leading to the well-defined pore walls and open porous network. The larger pores represent the lower crosslink density, whereas smaller pores represent the higher crosslink density.

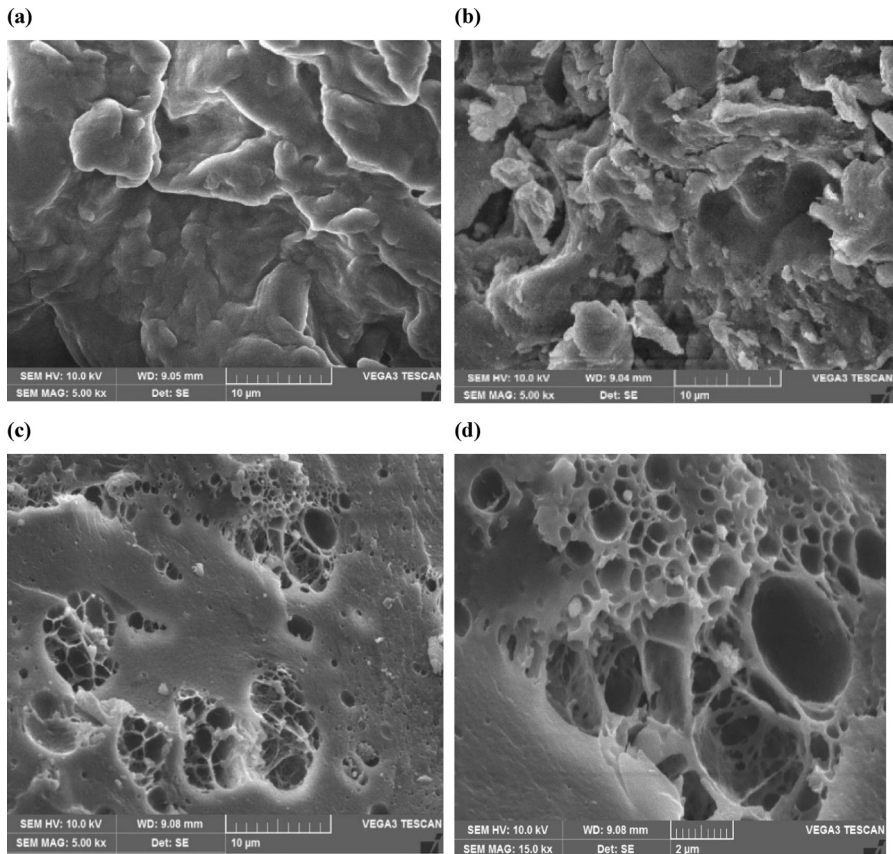


Fig. 5 SEM images of (a) Pure guar gum (b) Sodium alginate (c) and (d) GG/NaAlg-g-Am superabsorbent polymer at 10 μm and 2 μm respectively

Overall, the polymer network is mechanically stable with both mechanisms. These pores are believed to serve as areas for water permeation and as interaction sites for external stimuli with the hydrophilic groups of the graft copolymers [78–80].

Effect of pH on swelling

GG/NaAlg-g-Am Superabsorbent polymer possesses a 3D hydrophilic network, capable of holding an enormous amount of liquid compared to its dry weight. Swelling behaviour was evaluated for 48 h at every one-hour interval of time. From the Fig. 6, E.S. was analysed for acidic medium pH 4, neutral medium pH 7 and basic medium pH 10.2. In all three media, it was observed that there was a lag phase for the initial 5 h, followed by an accelerated log phase till 25–30 h, and then a stabilized or equilibrium phase till 48th hours. E.S. for basic medium (pH 10.2) was observed to be the lowest among the other two buffer media. Maximum E.S. attained for pH 10.2 is ~ 237 g/g. It indicates that a lesser concentration of H^+ ions creates a lesser protonation effect on anionic polysaccharide sodium alginate (COO^-) and acrylamide. Na^+ ions from NaOH shield the charges (counter-ion condensation), causing charge screening and reducing osmotic pressure. The imperfect electrostatic repulsion between anions results in a reduced osmotic pressure difference between the polymer network and the surrounding solution [81]. Hence charge shielding effect leads to a transformation in chain length. Moreover, excess hydroxide ions may lead to degradation or partial hydrolysis of the polymer backbone (especially amide bonds

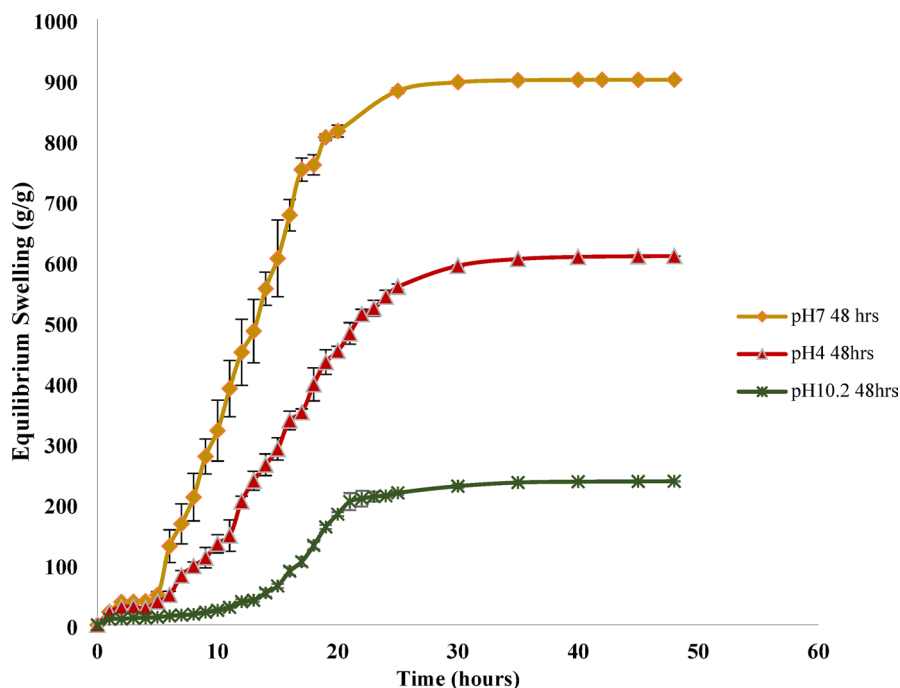


Fig. 6 Effect of pH (acidic, basic and neutral medium) on E.S

in polyacrylamide), reducing structural integrity and water uptake. Interchain ionic interactions or precipitation may occur, causing network collapse, which results in a lower crosslinking effect and acts as a barrier to enter the fluid in the void space, thus resulting in lower swelling equilibrium [82].

Based on the graph, the swelling behaviour of the GG/NaAlg-g-Am Superabsorbent polymer at pH 4 (acidic condition) reached an equilibrium swelling capacity of approximately 640 g/g, which is about 28.9% lower than the swelling observed at neutral pH (pH 7, ~ 900 g/g). The swelling curve shows a gradual increase during the initial 20 h, followed by a slower approach to equilibrium, indicating reduced network expansion due to protonation effects and enhanced intra-molecular hydrogen bonding under acidic conditions. The outcome may be attributed to the reduction of the aldehyde (C = O) and hydroxyl (-OH) groups in guar gum, which occurs due to the formation of hydrogen bonds across the double bond when a greater number of H⁺ ions are present in the acidic medium. The crosslinked polymer network is structured to retain water effectively. Under acidic conditions, the carboxylate groups (-COO⁻) in the polymer get protonated to -COOH, reducing the net negative charge on the hydrogel matrix. This causes a decrease in electrostatic repulsion, resulting in network contraction [83]. Excessive protonation leads to increased density within the polymer's crosslinked framework, resulting in the swelling capacity reaching a saturation point after the 100-minute mark [84, 85].

The hydrogel at pH 7 exhibits the highest swelling, reaching equilibrium at around 900 g/g within ~ 24 h and remaining constant thereafter. This can be elucidated as sodium alginate and guar gum exhibit the highest hydrophilicity at neutral pH due to the presence of an equilibrium between H⁺ and OH⁻ ions. Therefore, water retention capacity within the three-dimensional network increases. At neutral pH, the ionization of functional groups (-COOH, -OH, -NH₂) is moderate, allowing optimal hydrogen bonding and electrostatic repulsion between polymer chains. The network expansion is maximized due to balanced osmotic pressure from mobile counterions and polymer network elasticity [86, 87]. Once saturation between H⁺ and OH⁻ ions occurs, no hydrogen bonds from water molecules can be formed; therefore, constant E.S. is maintained [88].

Swelling kinetics

Swelling behaviour is an intrinsic property of a polymer. The kinetic study on reaction rate and its external affecting factors like concentration, time and pH plays a vital role in emphasizing the reactions of the polymer network. Given the observed swelling profile—characterized by an initial lag phase, followed by a rapid increase and a distinct equilibrium plateau—the Gompertz model is the most suitable for capturing the full kinetics of the system [89]. Additionally, Weibull and Logistic models have also been applied for sigmoidal swelling behaviour, with the Weibull model offering shape flexibility and the Logistic model being appropriate when the swelling curve is more symmetrical [90, 91]. In the context of the present study, polynomial regression modelling serves as a powerful empirical tool to describe and compare the swelling behaviour of hydrogels under varying pH conditions. Given the complex, non-linear, and time-dependent nature of hydrogel swelling, which typically involves an initial

lag phase, a rapid expansion phase, and eventual saturation, the use of a polynomial equation enables flexible curve fitting that captures all critical features of the swelling profile [92].

The characteristic equation and the regression coefficients across the three different media pH 4, pH 7, and pH 10.2, for the said models are tabulated in Table 1.

Polynomial regression model

From Fig. 7a at pH 7, the hydrogel exhibited the highest swelling capacity, with the equilibrium swelling reaching over 900 g/g. The swelling kinetics were best described by the fourth-degree polynomial Eq. 11, with a coefficient of determination $R^2=0.9911$, indicating a strong correlation with the experimental data. The equation captures the initial lag phase, followed by a rapid swelling period and a plateau, consistent with a typical sigmoidal swelling profile. The high swelling response at neutral pH suggests that the polymer network undergoes optimal ionization and electrostatic repulsion, facilitating enhanced chain relaxation and maximum water uptake. This condition reflects a balanced environment for charge interactions within the hydrogel matrix.

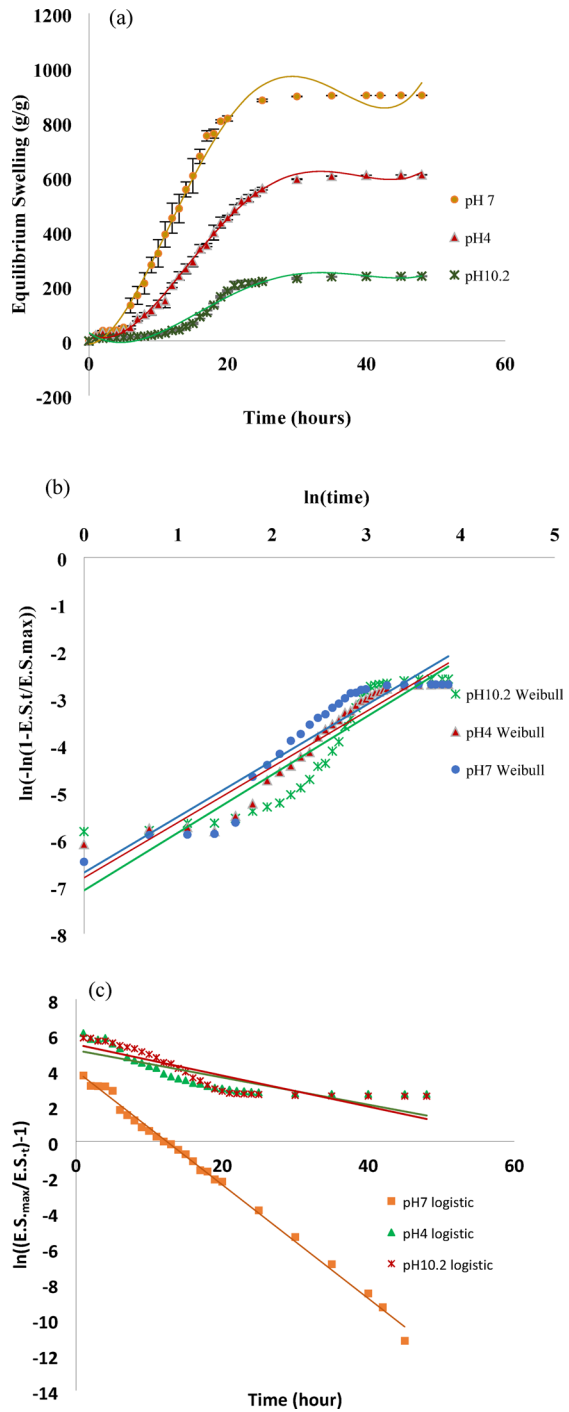
$$y = 0.002x^4 - 0.1959x^3 + 5.0706x^2 - 1.5882x - 11.258 \quad (11)$$

At pH 4, the swelling behaviour was moderate, with a lower maximum swelling (~ 600 g/g) compared to neutral pH. The corresponding polynomial Eq. 12 showed an excellent correlation with the data ($R^2=0.997$), the highest among all three pH conditions. The equation characterizes a relatively slower and more gradual swelling trend, indicative of suppressed ionization of functional groups such as carboxylates under acidic conditions. The likely protonation of ionizable groups ($-\text{COO}^-$ to $-\text{COOH}$) reduces electrostatic repulsion, resulting in a denser polymer network structure that restricts water diffusion and swelling. Despite the reduced magnitude, the predictability and consistency of the swelling behaviour under acidic conditions are well captured by the fitted model.

$$y = 0.0012x^4 - 0.1228x^3 + 3.7793x^2 - 14.778x + 27.702 \quad (12)$$

In alkaline conditions (pH 10.2), the hydrogel demonstrated the lowest swelling performance, with equilibrium swelling values not exceeding 250 g/g. The swelling kinetics were modelled by the polynomial Eq. 13, yielding a good fit with $R^2=0.9688$. This equation reflects a slow and limited swelling progression, likely due to excessive deprotonation of functional groups and possible ionic screening effects from hydroxyl ions or metal cations present at high pH. Such conditions may lead to a reduction in effective charge density and destabilization of the polymer network, resulting in minimal chain expansion. Additionally, hydrogel degradation or crosslink instability may further contribute to the restricted swelling observed in strongly basic environments. The fitted model thus offers valuable insight into the kinetic suppression of swelling under alkaline stress.

Fig. 7 Mathematical models illustrate swelling kinetics (a) polynomial regression, (b) Weibull, (c) Logistic and (d) Gompertz



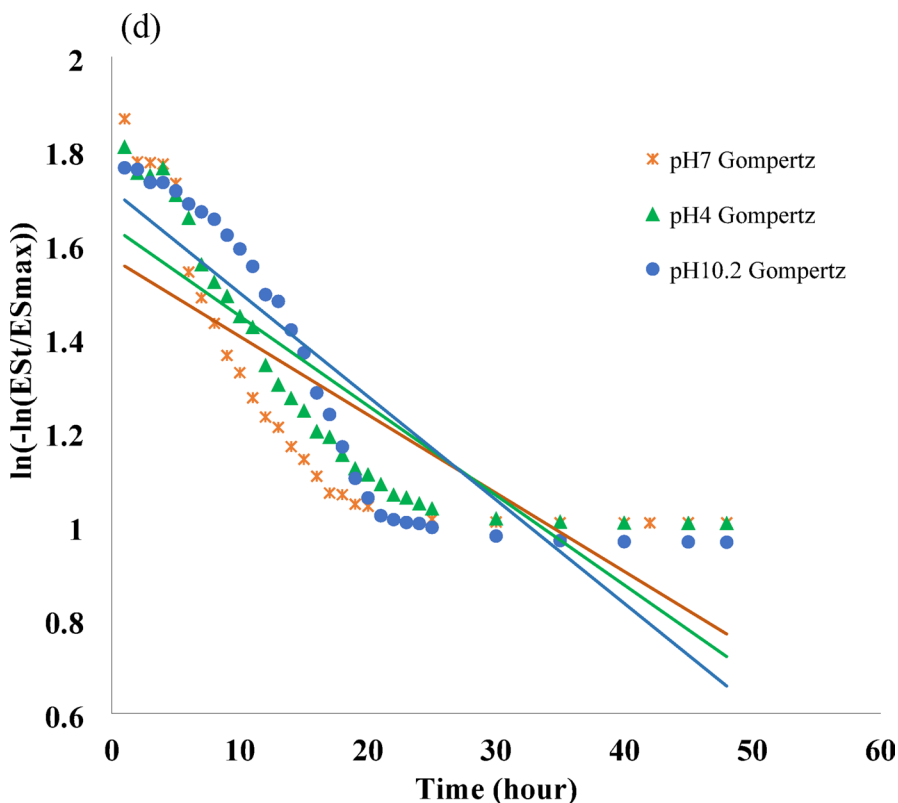


Fig. 7 (continued)

$$y = 0.0006x^4 - 0.0678x^3 + 2.2824x^2 - 16.859x + 29.985 \quad (13)$$

The polynomial equations allow for the calculation of critical points such as the maximum swelling rate (inflection point) and the approximate time to equilibrium, which are not easily extracted from raw data. This model thus enables direct visual and analytical comparison of swelling behavior across pH environments, revealing that swelling is most pronounced at neutral pH and significantly restricted under acidic and alkaline conditions—insights that would be difficult to achieve without such mathematical fitting.

Weibull model

The Weibull model, widely recognized for its ability to capture sigmoidal swelling kinetics with lag–log–plateau characteristics, was successfully applied to describe the swelling behaviour of the polymer matrix under varying pH conditions. From Fig. 7b at pH 10.2, the model exhibited a slope (shape parameter, b), $b > 1$ indicates sigmoidal swelling behaviour. Thus, $b = 1.2303$ and $a^2 = 0.8389$, indicating relatively fast swelling due to enhanced ionization and chain repulsion under alkaline conditions. At

pH 4, a slightly lower slope of 1.1881 with an R^2 of 0.8913 was observed, suggesting slower water uptake in the acidic medium due to protonation of functional groups limits network expansion. The best linear fit was obtained at pH 7, with a slope of 1.1793 and an R^2 of 0.9188, indicating a more balanced swelling profile that reflects simultaneous contributions from diffusion and polymer relaxation. These results confirm the suitability of the Weibull model across a range of pH values, particularly in capturing the asymmetry and transition kinetics typical of polymer swelling. Similar model applicability has been reported in hydrogel-based delivery systems and pH-sensitive matrices by Siepmann and Peppas [97], and Siegel and Rathbone [98], reinforcing its relevance for swelling studies in variable pH environments.

Logistic model

The logistic model, characterized by its symmetrical sigmoidal shape, is effective in describing swelling behaviours where the rate of water uptake increases rapidly after an initial lag phase and then slows down as equilibrium is approached. Mathematically, the model assumes equal curvature in the acceleration and deceleration phases, making it suitable for systems where swelling is governed by both polymer relaxation and diffusion processes in a balanced manner [99]. In the present study, Fig. 7c, regression analysis showed marked pH-dependent variations in model fit. At pH 7, the logistic model achieved the highest correlation ($R^2 = 0.995$), indicating that neutral conditions closely follow the idealized symmetrical swelling predicted by the logistic equation. This could be attributed to optimal ionization and network expansion at physiological pH, resulting in balanced water ingress and matrix relaxation. At pH 4, the model fit was moderate ($R^2 = 0.692$), reflecting deviations from symmetry due to suppressed ionization under acidic conditions, which delays swelling and causes a more pronounced lag phase. Similarly, at pH 10.2, the fit decreased ($R^2 = 0.756$), likely because strong alkalinity induces rapid initial swelling followed by a more gradual approach to equilibrium, thus introducing asymmetry that the logistic model cannot fully capture. Previous studies further support our findings. For instance, Slauter et al. [100] investigated the swelling behaviour of interpenetrating polymer networks using mathematical models, including the logistic model, and demonstrated its suitability for describing symmetrical swelling patterns. Similarly, Zhang et al. [101] applied the logistic model to analyse swelling kinetics in their work on bacterial cellulose/acrylic acid-based pH-responsive smart dressings prepared via graft copolymerization, highlighting the model's robustness in capturing near-symmetrical swelling behaviour. These reports reinforce our observation that the logistic model achieved its highest predictive accuracy under pH 7 conditions, where the swelling profile most closely followed a symmetrical sigmoidal pattern.

Gompertz model

The Gompertz model effectively described the swelling kinetics of the hydrogel system across all pH conditions, capturing the characteristic lag–log–plateau behaviour. From Fig. 7d at pH 7, the swelling progression was more gradual and symmetrical, indicating balanced polymer chain relaxation and solvent penetration, whereas at pH 4 and pH 10.2, faster initial swelling was followed by an earlier deviation from linearity, suggesting pH-induced modulation of network expansion. The regression coefficients for pH 4, pH 7, and pH 10.2 were 0.745, 0.624, and 0.780, respectively, indicating the best model fit under alkaline conditions, followed by acidic conditions, with neutral pH showing a slightly lower fit due to its uniform swelling profile. This trend is consistent with Farmanborder et al., [102] research study on pH-Sensitive silica-based core–shell nanogel prepared via RAFT polymerization to investigate the core size effect on the release profile of doxorubicin, where the Gompertz model demonstrated high predictive capability in capturing asymmetric swelling kinetics influenced by ionic strength and electrostatic interactions.

Agricultural application

Biodegradability studies

The biodegradability test of the synthesized GG/NaAlg-g-Am superabsorbent polymer is carried out by the soil burial method. SEM morphology in Fig. 8a confirms the polymer degradation due to reduced projections and deformation of the clear porous polymer network.

The Fig. 8b depicts the decay curve in which the relationship between the time required to degrade the amount of polymer is illustrated. Half-life, i.e., time required to degrade 50% of the initial weight, is obtained as 110 days. Thus, the rate of degradation of the synthesized polymer is considered moderate. From the plot with an exponential trendline, the degradation rate constant, k , is found to be 0.007. The cal-

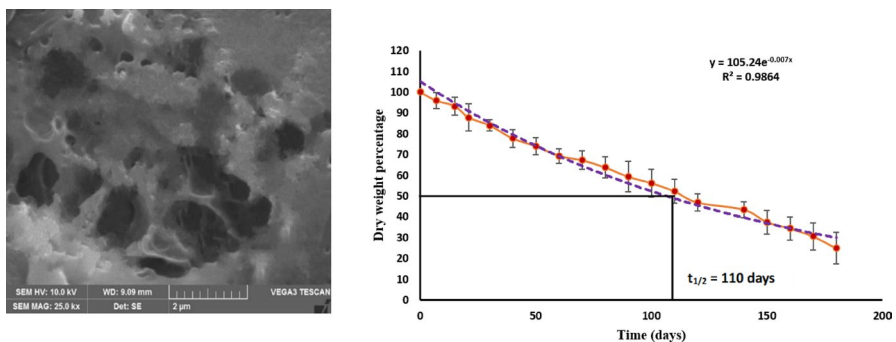


Fig. 8 (a) SEM morphology of GG-NaAlg-g-Am polymer on the 80th day of biodegradability studies. (b) Osmon's model plot representing the half-life period

culated degradation rate constant, k , from Eq. (6) is obtained as 0.0065. This confirms that degradation follows Olson's model.

Initially, the rate of degradation is observed to be faster till 100 days, when about 49% of degradation occurred, followed by a slower phase, in which the next 40 days of degradation of only 13% is observed. Next was the accelerated phase, wherein 19% of degradation is attained in 40 days.

The highest percentage of decay in the initial phase may be ascribed to abundant oxygen availability due to the hydrophilicity of guar gum and sodium alginate. Compounds that are susceptible to a higher rate of water absorption exhibit faster degradation. The slower phase of degradation can be explained due to the higher water-to-polymer ratio results in anaerobic conditions. The larger amount of water in the superabsorbent polymer after saturation decreases the oxygen content in turn inhibits microbial action, which decreases the decay rate. The last phase of degradation is accelerated due to smaller polymer chains that tend to degrade faster than longer chains [103, 104].

Effect of GG/NaAlg-g-Am superabsorbent polymer on soil particle density

Soil particle density serves as an indicator of the available space for root development, air circulation, and water retention. The data presented in Fig. 9 illustrates that as the percentage of GG/NaAlg-g-Am superabsorbent polymer treatment in the soil increases, the particle density of the soil decreases. Notably, there is a significant reduction in particle density from the control measurement of 2.69 g/mL to 2.41 g/

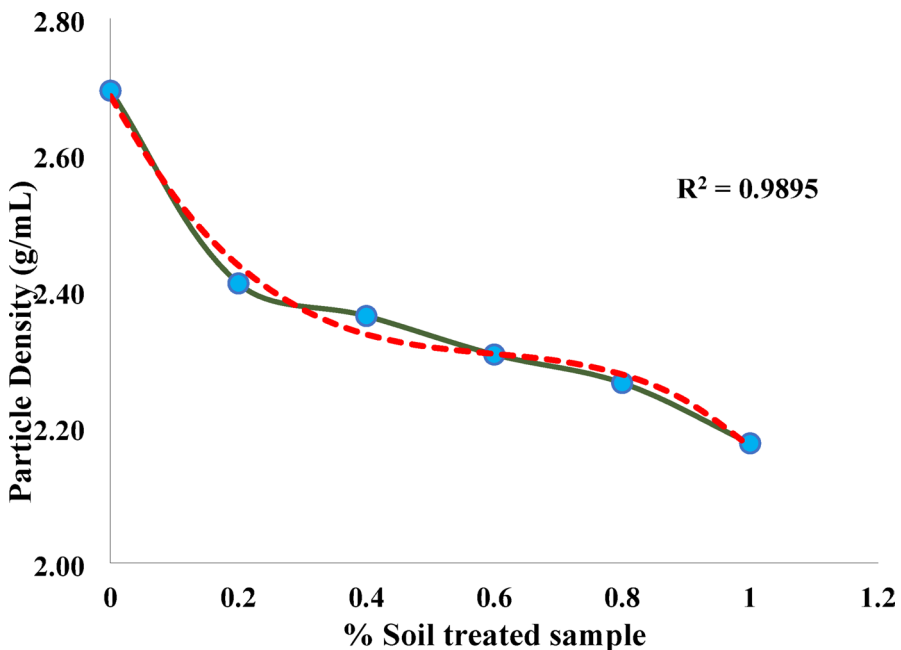


Fig. 9 Effect of GG/NaAlg-g-Am superabsorbent polymer on soil particle density

mL at a 0.2% PTS sample, followed by a gradual decline in particle density corresponding to the treatment percentage. This outcome can be attributed to the polymer treatment's influence on the soil, which enhances its organic content, thereby reducing particle density [105, 106]. The relationship between particle density and %PTS, characterized by a regression coefficient of 0.989, follows a third-order reaction as represented by the corresponding Eq. 14. This indicates that the reaction rate is directly proportional to the concentration of three reacting molecules, suggesting that particle density is directly proportional to the three reactant molecules of the % polymer-treated soil sample. Sarmah et al.. have made comparable observations, indicating that the percentage of soil density diminishes as the hydrogel content increases [107].

$$y = -1.5509x^3 + 2.7712x^2 - 1.7357x + 2.6861 \quad (14)$$

Effect of GG/NaAlg-g-Am superabsorbent polymer on soil porosity

Soil porosity is defined as the proportion of the overall volume of soil that is occupied by pore spaces. The Fig. 10 illustrates that the porosity of soil without GG/NaAlg-g-Am superabsorbent polymer treatment was observed to be 42% but 0.2% treated soil exhibited 52.3% porosity. There was a gradual increase in porosity percentage till 55.4% was observed for 1% PTS. The relationship between the %porosity and the % PTS is described by Eq. 15. This result can be explained as when hydrogel absorbs water due to its three-dimensional network space, it swells, resulting in the formation

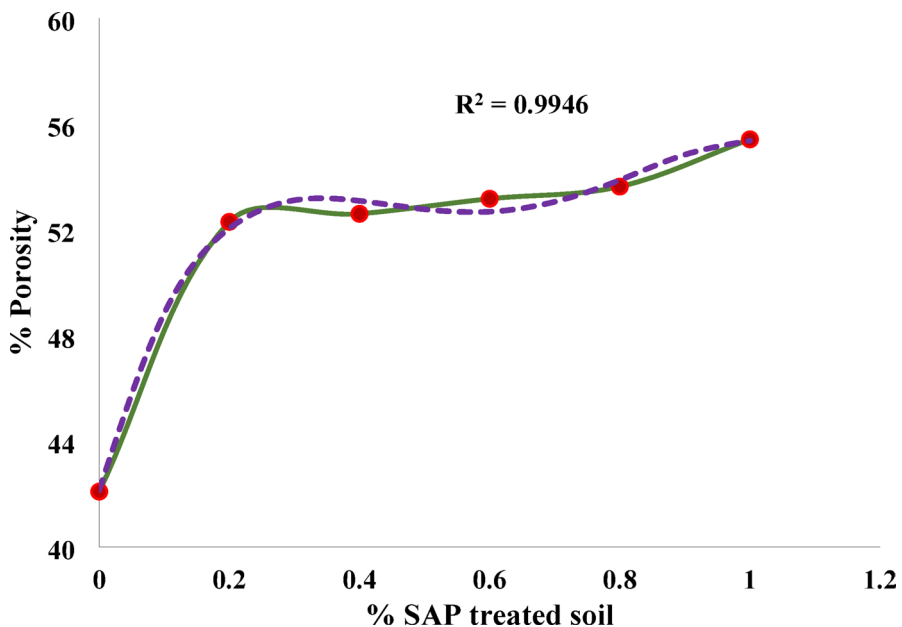


Fig. 10 Effect of GG/NaAlg-g-Am superabsorbent polymer on soil porosity

of gaps between soil particles, which in turn enhances the overall porosity of the soil. Similar observations have been recorded by Sahar et al., where gelatine and carboxy methyl cellulose have been used as hydrogels in which soil porosity increases with an increase in hydrogel percentage [108].

$$y = -113.71x^4 + 291.09x^3 - 253.72x^2 + 89.565x + 42.142 \quad (15)$$

Effect of GG/NaAlg-g-Am superabsorbent polymer on the maximum water holding capacity

The experimental results demonstrated in Fig. 11 that the control soil had a maximum water holding capacity (MWHC) of 29%, meaning 100 g of untreated air-dried soil retained 29 g of water at saturation. Upon treatment with 0.2% PTS, the MWHC increased markedly to 59%, representing a 103% enhancement compared to the control. This substantial improvement was attributed to the swelling behaviour of the GG/NaAlg-g-Am superabsorbent polymer, which formed a three-dimensional cross-linked network capable of absorbing and retaining significant volumes of water within its matrix. Superabsorbent polymers typically retained several hundred to over 1000 times their weight in water, owing to their hydrophilic functional groups ($-\text{OH}$, $-\text{COOH}$, $-\text{NH}_2$) and covalent crosslinks introduced by N, N'-Methylenebisacrylamide. Mechanistically, as the polymer swelled in the soil, it formed interconnected porous structures that trapped water molecules and reduced leaching and evaporation losses, thereby enhancing soil moisture retention. Quantitatively, for every 0.2% increment in polymer dosage, a 5–7% rise in MWHC was observed, indicating a consistent, dose-dependent response. At 1% PTS, the MWHC reached 83%, reflecting a 186% increase compared to control. This linear trend supported the conclusion that the improvement in water holding capacity was directly proportional to the polymer concentration, validating the effectiveness of the GG/NaAlg-g-Am superabsorbent polymer as a soil amendment for improving water retention and mitigating drought stress [109, 110]. Nazeri et al. reported comparable findings, utilizing a carboxymethylcellulose-grafted poly(Acrylic Acid-Co-Vinyl Acetate) hydrogel as a slow-release fertilizer. Their study demonstrated an increase of 11.8% in moisture holding capac-

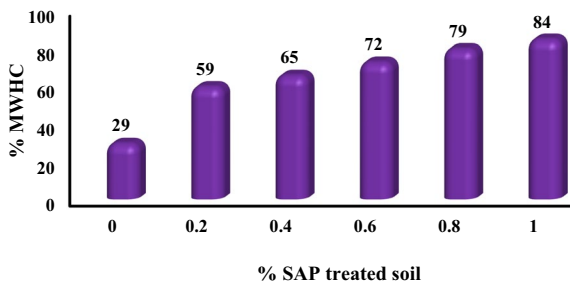


Fig. 11 Effect of GG/NaAlg-g-Am superabsorbent polymer on Maximum Water Holding Capacity

ity (MWHC) when comparing the controlled soil sample to the soil sample treated with 2% hydrogel [111].

Effect of SAP on water retention capacity of soil

Figure 12a and b compared the water retention capacity between the control soil and GG/NaAlg-g-Am superabsorbent polymer-treated soil. At saturation, i.e., at zero bar pressure, the water retention capacity of untreated soil was recorded as 27.58%, while a significant increase to 44.94% was observed with 0.2% polymer-treated soil. There was an approximate 6% to 7% rise in water retention with every 0.2% increment in GG/NaAlg-g-Am polymer treatment. As the pressure increased to 0.5, 1, 2, 5, and 10 bar, the moisture retention percentage correspondingly decreased. The highest moisture retention of 64.19% observed for 1% GG/NaAlg-g-Am polymer-treated soil at zero bar pressure, compared to 27.58% in the control soil, reflecting the efficient water absorption and retention capability of the superabsorbent hydrogel. The 132% increase in water retention is due to the hydrophilic nature and porous structure of the crosslinked GG/NaAlg-g-Am polymer network, which forms capillary-like microchannels within the soil, enhancing water retention via physical entrapment and hydrogen bonding. At zero bar pressure, capillary forces dominate and the hydrogel is fully swollen, maximizing water retention. However, when the pressure is increased to 10 bar, external pressure exceeds capillary and osmotic forces, causing the polymer matrix to deswell and release water. As a result, the moisture retention decreases to 22.39%, marking an 186% reduction compared to zero bar. This behaviour is characteristic of pressure-sensitive hydrogel systems, where interconnected pores collapse under pressure, releasing entrapped water [112]. Additionally, crosslinking density and polymer–soil interactions influence this response. Highly crosslinked networks resist pressure-induced collapse better, maintaining sustainable swelling and moisture retention. However, excessive pressure compresses both the polymer and soil matrix, reducing pore volume and water-holding capacity [113].

Thus, results can be concluded by the principle that soil water retention is primarily governed by capillary action between soil particles. Water molecules were held to soil particles via adhesive forces, and with increasing pressure, the pore space

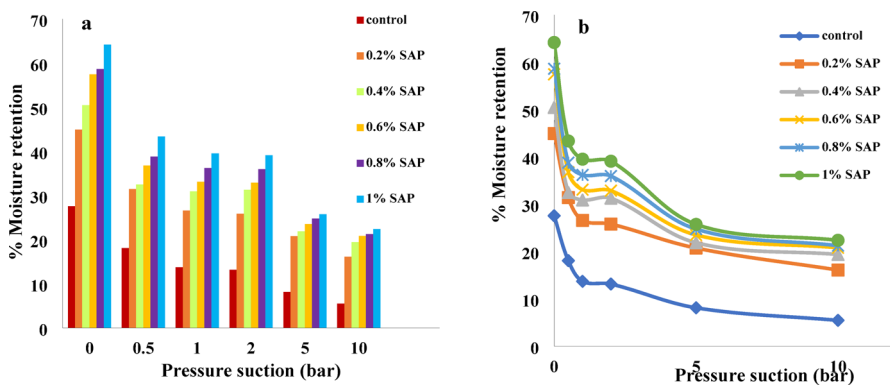


Fig. 12 Effect of GG/NaAlg-g-Am superabsorbent polymer on water retention capacity

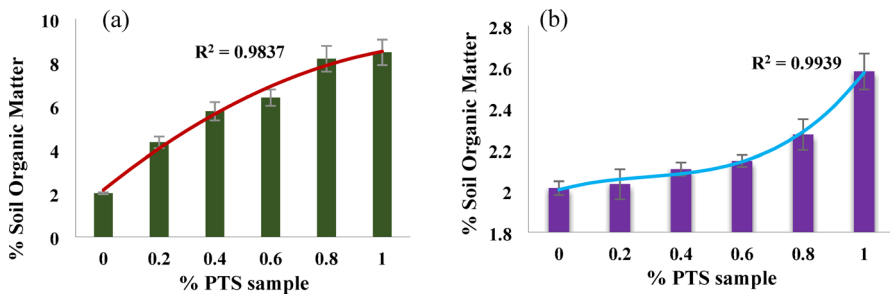


Fig. 13 Effect of GG/NaAlg-g-Am superabsorbent polymer on % Soil Organic Matter (a) 7th day of polymer treatment (b) 150th day of polymer treatment

between soil particles diminished, reducing the soil's capacity to retain water. Consequently, water retention percentage declined with increasing pressure in both PTS and control soil samples [114]. In the case of polymer treatment of the soil, it was proven that polymer treatment has a high impact on the water retention percentage due to the porosity of the soil increases due to the polymer network space. An identical trend is observed that water retention was increased with an increase in the percentage of polymer treatment, even in the case of pressure increase from zero bar to 10 bar [115]. Similar observations are noted by Albalasmeh et al., in which the corn growth was boosted by the application of hydrogel in the soil [116].

Effect of SAP on soil organic matter

Soil Organic Matter (SOM) serves as a crucial indicator for assessing soil quality. An increase in SOM correlates positively with enhanced plant growth potential. The two graphs, Fig. 13 (a), (b), demonstrate the short-term (7th day) and long-term (150th day) impact of GG/NaAlg-g-Am superabsorbent polymer (SAP) treatment on soil organic matter (SOM). At day 7, SOM increased sharply from ~2% (control) to ~8.5% at 1% SAP, indicating a 325% relative increase. This immediate elevation can be attributed largely to the inclusion of undecomposed polymer residues in SOM measurements, particularly when using Walkley-Black or Loss on Ignition (LOI) methods, which measure total organic content without differentiating between stable humus and added organic polymer. The polymer's hydrophilic biopolymer backbone (guar gum and sodium alginate) contributes directly to this SOM rise through its organic load and high-water retention, which enhances moisture availability and promotes initial microbial proliferation.

In contrast, at day 150, SOM increased more modestly from ~1.88% to ~2.7%, reflecting a 43% enhancement over control. The quadratic Eq. 16 for the 7th day indicates a rapid increase in SOM with increasing SAP dosage, showing a strong fit ($R^2 = 0.983$) and suggesting initial high SOM accumulation due to added polymer content. The cubic Eq. 17 for the 150th day reflects a gradual and stabilized increase in SOM over time, with a very high fit ($R^2 = 0.994$) indicating sustained organic matter formation from polymer biodegradation and soil carbon stabilization. This result signifies the true stabilization and humification of organic matter derived from the biodegradation of the polymer in soil. Over time, microbial activity facilitated the enzymatic

breakdown of guar gum and sodium alginate into simpler organic molecules, which were incorporated into stable humic fractions. Additionally, covalently crosslinked polyacrylamide chains (via N, N'-Methylenebisacrylamide) degraded slowly, contributing to long-term soil organic carbon pools. The hydrophilic functional groups ($-\text{OH}$, $-\text{COOH}$, $-\text{NH}_2$) formed organo-mineral complexes that aided in SOM stabilization. Furthermore, the porous hydrogel matrix created microenvironments that improved microbial colonization and reduced nutrient leaching, further supporting sustainable SOM accumulation [117]. Mechanistically, the initial spike in SOM was a physical addition of organic material, while the long-term increase reflects a biologically driven transformation of that material into functionally beneficial, stabilized organic matter. The results collectively confirm that GG/NaAlg-g-Am SAP serves both as a short-term organic amendment and a long-term soil conditioner, enhancing soil carbon content sustainably through controlled biodegradation and effective water management. Their degradation contributes to the enhancement of organic matter in the treated soil, which in turn may lead to improved nutrient availability for plants [118, 119].

The trendline Eqs. 13 and 14 are for 7th day %SOM and 150th day %SOM, respectively.

$$y = -0.1561x^2 + 2.3636x - 0.0569 \quad (16)$$

$$y = 0.0092x^3 - 0.0671x^2 + 0.1873x + 1.8763 \quad (17)$$

Conclusion

The study underscored the importance of hydrogel treatment in enhancing soil properties. The synthesized GG/NaAlg-g-Am superabsorbent polymer demonstrated significant potential as an agricultural amendment. This polymer was created through microwave irradiation graft polymerization, with careful optimization of various reaction parameters, including the concentrations of acrylamide, ammonium persulfate, and N, N'-Methylenebisacrylamide, as well as microwave power and reaction duration, resulting in a maximum grafting percentage of 393%. The influence of pH on the swelling capacity of the GG/NaAlg-g-Am superabsorbent polymer was thoroughly examined. The findings revealed that the GG/NaAlg-g-Am superabsorbent polymer exhibited the highest E.S. of 900 g/g at pH 7, with comparatively lower swelling capacities of 607 g/g at pH 4 and 229 g/g at pH 10.4, indicating optimal performance under neutral conditions. Importantly, the polymer demonstrated sustainable swelling behaviour, characterized by gradual and controlled water uptake over 48 h without structural collapse. The combined swelling performance and structural integrity underscore the potential of GG/NaAlg-g-Am superabsorbent polymer as an effective and technically advanced soil amendment material. Swelling kinetics were analysed using various models, including Korsmeyer-Peppas's, Higuchi, Zero order, First order, Schott's pseudo-second-order, and Hixson-Crowell models. The results indicated that the swelling kinetics were primarily governed by diffusion, accompa-

nied by polymer erosion due to stress from fluid penetration. A 1% PTS treatment resulted in a 31.67% increase in soil porosity and a 23.48% rise in bulk density, indicating improved soil aeration and structural stability. These modifications are beneficial for root penetration, water infiltration, and overall soil structure. Furthermore, the maximum water-holding capacity increased by 189% in PTS relative to the control soil. Although water retention decreased with increasing pressure, soils treated with higher polymer concentrations consistently exhibited significantly greater moisture retention than untreated soils, demonstrating the effectiveness of the polymer in enhancing soil water retention under varying conditions. The enhanced maximum water holding capacity and water retention are critical factors for improving crop yields in drought-prone agricultural areas. GG/NaAlg-g-Am superabsorbent polymer treatment increased SOM from 1.99% (control) to 8.5% on day 7, primarily due to undegraded polymer residue. By day 150, SOM rose from 1.88% to 2.7% at 1% polymer treatment, reflecting a 43% long-term enhancement attributed to gradual polymer biodegradation, carbon stabilization, and improved soil structure. These results demonstrate the efficacy of the polymer in both immediate and sustained enrichment of soil organic matter. Soil organic matter serves as a crucial indicator of soil fertility and quality; thus, a higher organic matter content correlates with improved soil health, which is essential for optimal plant production. The biodegradability of the GG/NaAlg-g-Am superabsorbent polymer aligns with the Oslon model, exhibiting a degradation rate constant of 0.0065 and a half-life of 110 days. This indicates that the synthesized polymer is environmentally friendly due to its biodegradable nature. The analysis suggests that the GG/NaAlg-g-Am superabsorbent polymer has significant potential as a soil amendment.

Author contributions S B the first author conducted the experiments and wrote the manuscript draft. G N, RRN and Pratik analyzed the data obtained and suggested further protocols. They helped in scrutinizing the plotted graphs and the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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